

Final Project Report of ECON 425

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Credit Default Prediction Using Logistic Regression and XGBoost

I . Problem Identification

Research Question

The primary objective of this study is to develop a credit risk model that can effectively predict whether a credit card holder will default on their payment next month. Credit risk modeling is important in the financial industry as it helps banks and financial institutions determine the likelihood of a customer defaulting on a loan or credit card payment. By predicting defaults, banks can:

1. Optimize risk management strategies and reduce losses.
2. Make data-driven lending decisions, adjusting credit limits or interest rates based on predicted risk.
3. Improve regulatory compliance by ensuring responsible lending practices.

Choice of Data

The dataset used in this study is the Taiwan credit card default dataset, which contains 30,000 credit cardholders' records, including demographic information, credit history, and repayment behavior. The dataset is well-suited for this analysis as it includes a wide range of explanatory variables that can help predict default behavior.

- **LIMIT_BAL**: Credit limit (NTD), including personal consumption credit and supplementary credit for family.
- **SEX**: Gender, where:
 - 1 = Male
 - 2 = Female
- **EDUCATION**: Education level, where:
 - 1 = Graduate school
 - 2 = University
 - 3 = High school
 - 4 = Others (including unknown, etc.)
- **MARRIAGE**: Marital status, where:
 - 1 = Married
 - 2 = Single
 - 3 = Others (including unknown status)
- **AGE**: Age (years).
- **PAY_0 - PAY_6**: Historical repayment records (April to September 2005):
 - -1 = Paid on time
 - 1 = Payment delayed by 1 month
 - 2 = Payment delayed by 2 months
 - ...
 - 9 = Payment delayed by 9 months or more
- **BILL_AMT1 - BILL_AMT6**: Bill amounts (NTD) for April to September 2005.
- **PAY_AMT1 - PAY_AMT6**: Payment amounts (NTD) for April to September 2005.
- **default_payment_next_month**: Whether there is a credit card default in the next month, where:
 - 1 = Default
 - 0 = No default

Fig 1. Raw data and field descriptions

II . Application of Machine Learning Tools

Model Selection

The target variable is binary (default or no default), making **classification** the appropriate choice. Regression models, which predict continuous values, are less suitable in this context because financial institutions require clear categorical predictions to make lending decisions, such as whether to approve or reject a credit application. Therefore, I compare two machine learning models:

1. **Logistic Regression:** A traditional, interpretable model commonly used in finance for credit scoring.
2. **XGBoost:** A powerful ensemble learning method that builds multiple decision trees to enhance predictive accuracy.

Each model has distinct **strengths**:

1. **Logistic Regression:** Known for its simplicity and transparency, it provides easily interpretable coefficients, making it ideal for regulatory compliance and financial decision-making.
2. **XGBoost:** A high-performing machine learning algorithm that leverages gradient boosting to capture complex nonlinear relationships in data. It is particularly useful when dealing with intricate interactions between multiple features and can uncover hidden patterns that simpler models might overlook, potentially improving predictive accuracy in diverse datasets.

Data Preprocessing

A key aspect of this study was ensuring proper **data preprocessing** before applying these models. Since different machine learning models have distinct data requirements, I carefully applied appropriate transformations to optimize their performance. To be more specific:

1. **One-Hot Encoding (For Logistic Regression):** Logistic Regression requires categorical variables to be represented numerically. To ensure proper interpretation, categorical features (e.g., Education, Marriage) were converted into multiple binary (0/1) variables using one-hot encoding. XGBoost does not require this step as it can handle categorical variables natively.
2. **Standardization (For Logistic Regression):** Logistic Regression is sensitive to feature scales, meaning variables with different ranges can disproportionately influence the model. Features such as bill amounts and previous payments vary significantly in magnitude, requiring standardization to bring them onto a similar scale. XGBoost, being a tree-based model, does not require standardization, as it naturally handles features of different scales.
3. **SMOTE (For Logistic Regression and XGBoost):** The dataset is highly imbalanced, with significantly fewer default cases compared to non-default cases. SMOTE (Synthetic Minority Over-sampling Technique) was applied to generate synthetic samples for the minority class, ensuring that the models are not biased toward predicting non-defaults. Both Logistic Regression and XGBoost benefit from this adjustment, as it improves recall by making the models more effective at identifying potential defaulters.

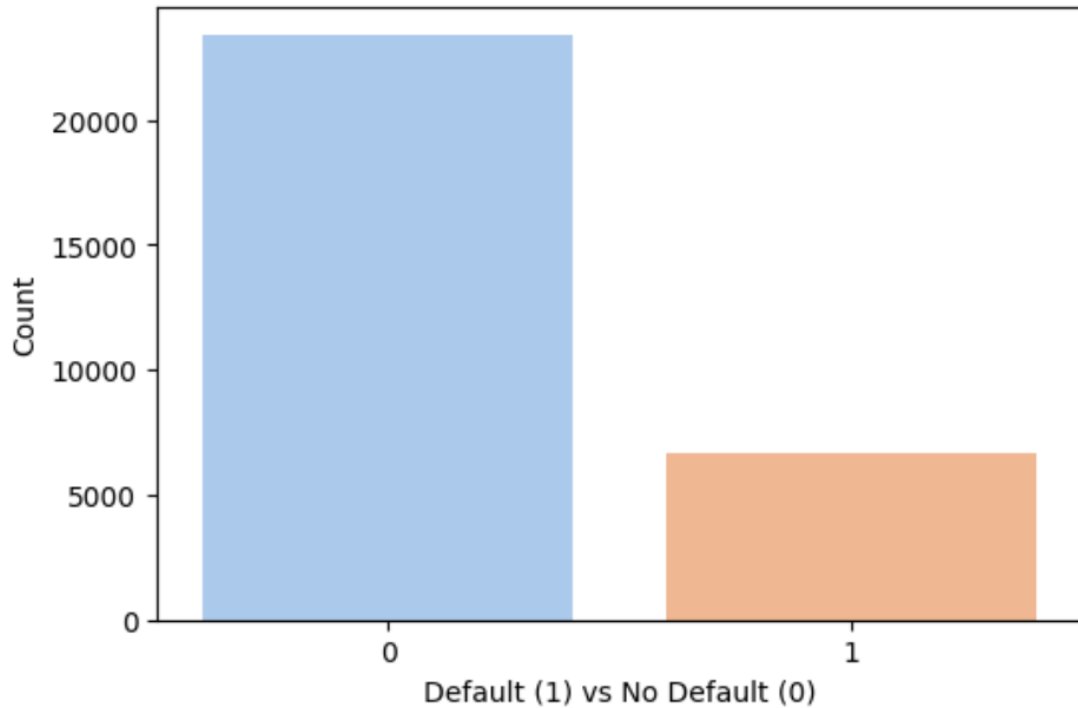


Fig 2. Distribution of Default Payment Next Month

Model	One-Hot Encoding	Standardization	SMOTE
Logistic Regression	√	√	√
XGBoost	×	×	√

Table 1. data preprocessing of two models

Model Performance Metrics

I evaluate the models based on three key metrics:

1. **Recall (for default class = 1)** – This is the **most critical metric** for risk management, as banks prioritize identifying potential defaulters over minimizing false positives. Missing a true defaulter (false negative) can lead to **significant financial losses**, as the bank extends credit to high-risk individuals who may fail to repay. The cost of a false negative is the total unpaid loan or credit amount, which can accumulate over time and result in bad debt write-offs.
2. **F1 Score (for default class = 1)** – This metric balances **precision and recall**, ensuring that the model does not **overly misclassify non-defaulters as defaulters** (false positives). While high recall is essential, excessive false positives could result in **unnecessarily rejecting creditworthy customers** or assigning them **higher interest rates**, potentially reducing customer satisfaction and bank revenue. The cost of a false positive is **lost business opportunities** and potential damage to the bank's reputation due to unfair lending decisions.

3. **ROC AUC (overall classification performance)** – Measures the model's **ability to distinguish between defaulters and non-defaulters** across different probability thresholds. A higher ROC AUC indicates a more effective model at ranking customers by risk level, allowing banks to make more informed, probability-based credit decisions.

Model	Recall (Default)	F1 Score (Default)	ROC AUC
Logistic Regression	0.633	0.458	0.656
XGBoost	0.475	0.476	0.664

Table 2. Model Performance of two models

The confusion matrices point the distribution of false positives and false negatives.

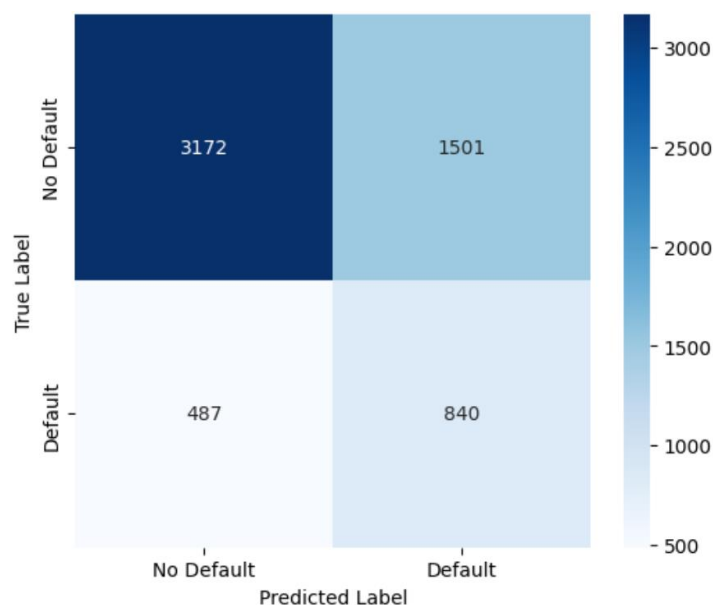


Fig 3. Confusion Matrix (Logistic Regression)

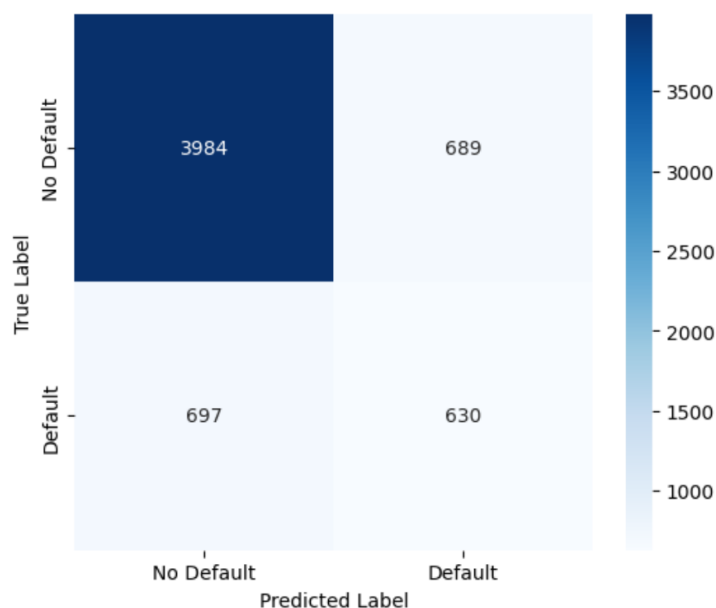


Fig 4. Confusion Matrix (XGBoost)

Key Observations

1. Logistic Regression outperforms XGBoost in Recall (0.633 vs. 0.475). This is critical because in credit risk assessment, banks want to catch as many defaulters as possible.
2. Two models have similar F1 score and ROC AUC.

Based on these results, I recommend using Logistic Regression for credit risk modeling. While this may seem counterintuitive—since XGBoost is often expected to outperform traditional models—there are specific reasons why Logistic Regression is better suited for this task, which I will explain in detail next.

III. Why Logistic Regression Performs Better in Credit Default Prediction

1. Financial Institutions Prioritize **Interpretability**: In credit risk modeling, financial institutions must justify decisions to regulators and customers. Logistic Regression provides clear coefficients that directly explain how each variable affects default probability. XGBoost, while powerful, is a "black-box" model, making it difficult to interpret why certain decisions were made.
2. **Recent Repayment Behavior is the Strongest Predictor of Default**. Feature importance analysis in Logistic Regression reveals that PAY_0 (most recent repayment status) is the most influential factor, with an absolute coefficient value of 0.5749. This makes sense intuitively—if a borrower has already delayed payments, they are more likely to default in the next month. XGBoost's feature importance ranking is less aligned with this financial intuition, giving higher importance to LIMIT_BAL (credit limit) rather than recent repayment patterns.
3. Logistic Regression Focuses on **Recall**: In credit risk, missing a defaulter (False Negative) is riskier than flagging a non-defaulter as risky (False Positive). Logistic Regression's higher recall (0.633 vs. 0.475) means fewer high-risk customers are misclassified as low risk. XGBoost prioritizes overall accuracy and AUC, which does not align perfectly with the objective of capturing the most defaulters. As seen in the confusion matrices, Logistic Regression correctly identifies **840** default cases (true positives) but misclassifies **487** defaulters as non-defaulters (false negatives). In contrast, XGBoost correctly identifies only **630** default cases but misclassifies **697** defaulters as non-defaulters. This demonstrates that Logistic Regression achieves higher recall in detecting defaulters, which is crucial for credit risk management.
4. **Stability & Robustness**: Logistic Regression is less sensitive to noise and overfitting, making it more reliable in real-world banking applications. XGBoost, as a tree-based model, is more prone to overfitting unless extensively tuned.

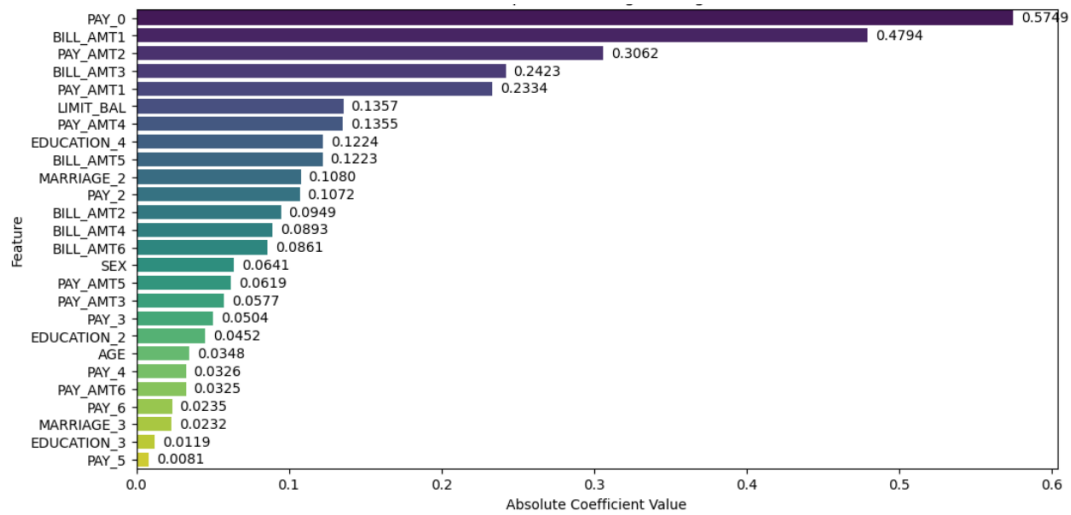


Fig 5. Feature Importances (Logistic Regression)

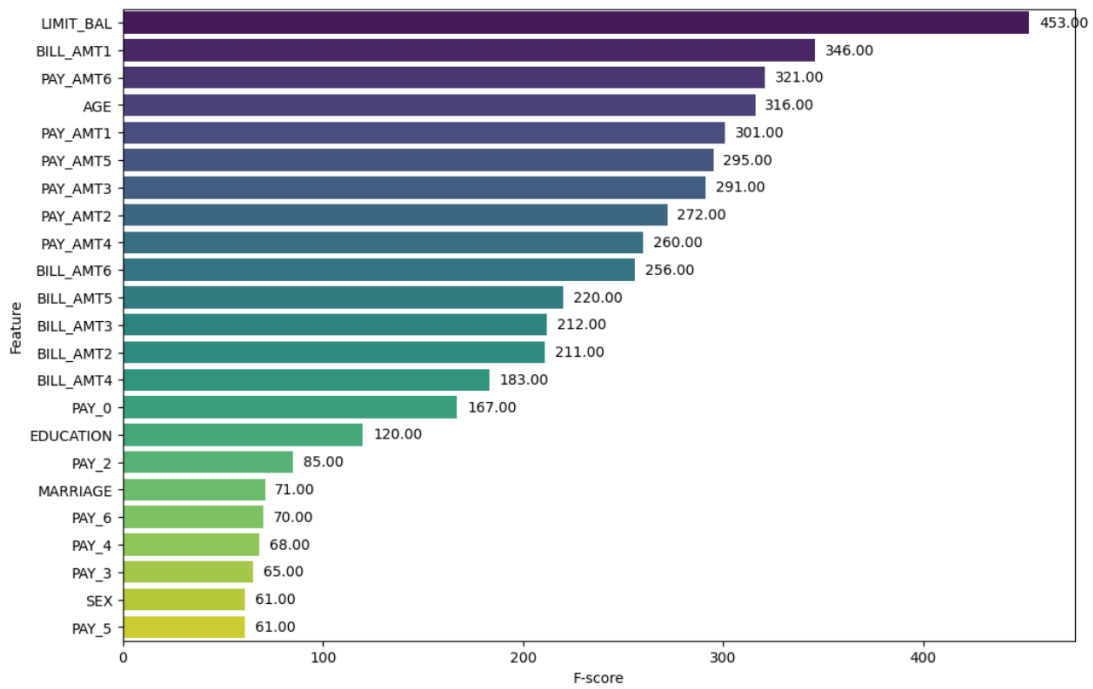


Fig 6. Feature Importance (F-score from XGBoost)

IV. Conclusion

Based on the analysis, Logistic Regression proves to be the better choice for credit risk modeling. Its higher recall ensures that more potential defaulters are identified, reducing financial risk for banks. Additionally, its stability and interpretability make it well-suited for regulatory compliance and real-world banking applications. While XGBoost remains a powerful tool, its advantages in handling complex nonlinear data are less relevant in structured financial datasets like credit risk assessment.

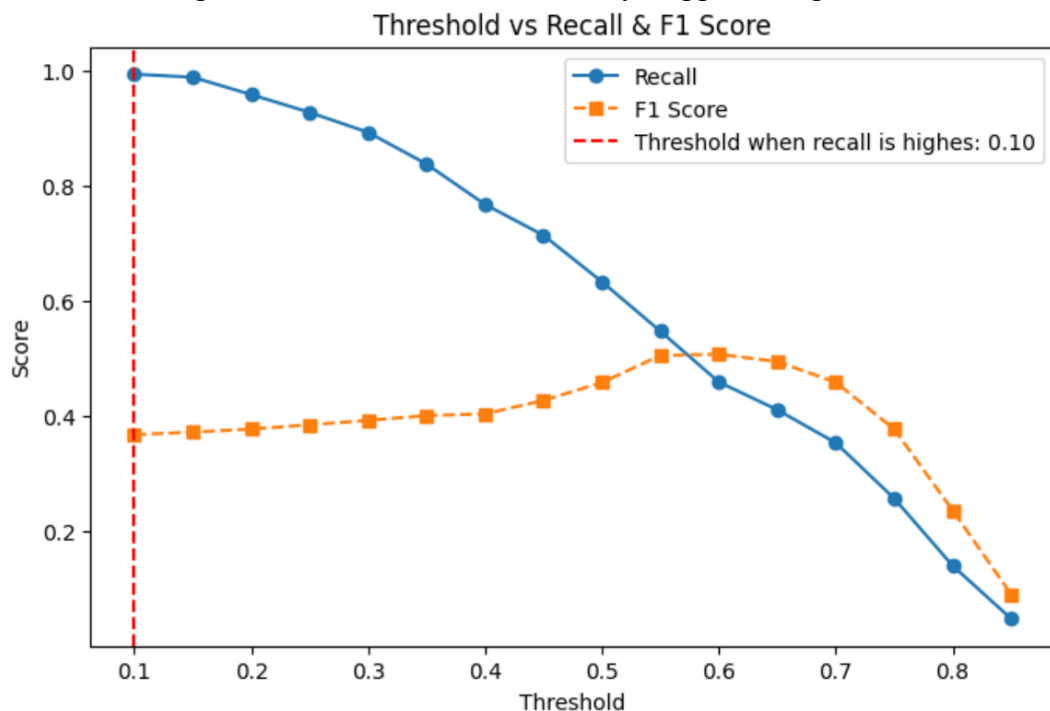
However, while **Logistic Regression performs well in this study, there are several limitations** to consider:

1. Limited Model Comparison:

This study only compares Logistic Regression and XGBoost. While these are widely used, other models such as Random Forest, K-Nearest Neighbors (KNN), or Neural Networks could be explored. Random Forest could provide a balance between interpretability and predictive performance, while KNN might offer insights into similarity-based classification.

2. Threshold Optimization in Logistic Regression:

The default classification threshold (0.5) may not be optimal for credit risk assessment. Adjusting the decision threshold (e.g., lowering it to increase recall) could improve default detection. However, this also increased false positives, meaning more customers were incorrectly flagged as high-risk.



Best Threshold for Maximum Recall: 0.10
Best Recall at this Threshold: 0.9932
F1 Score at this Threshold: 0.3674

Fig 7. Exploratory Threshold Analysis for Optimizing Recall in Logistic Regression

This trade-off is critical in credit scoring, where missing a defaulter is more costly than mistakenly rejecting a low-risk borrower. My results show that adjusting the threshold leads to a significant improvement in recall, but at the expense of precision. Future improvements could involve optimizing the threshold based on **business-defined cost functions** rather than just maximizing recall.

3. Feature Engineering & Alternative Data Sources:

The dataset is limited to demographics and financial history, but alternative data sources such as transaction history, employment stability, and macroeconomic factors could improve predictive power.